MEASLES IN THE CHILDREN'S EMERGENCY

1. CLINICAL PRESENTATION

1.1 MEASLES

- Highly contagious viral illness transmitted via airborne spread, respiratory droplets, and contact (direct person to person contact or direct contact with contaminated surfaces)
- Patients are infectious 4 days before & after onset of rash (Rash ± 4 days)
- The 4 stages of infection:
  - Incubation — 7 to 21 days
  - Prodromal — Fever
Triple Cs (coryza, cough, & conjunctivitis)
Koplik spots (onset **48h** before rash; disappears by **D2** of rash; lesions start to slough with rash onset)
  - Exanthem — Erythematous, maculopapular, blanching rash
Cephalocaudal & centrifugal spread
(head → neck → upper trunk → lower trunk → extremities)

    **Characteristic rash may not develop in immunocompromised patients and individuals with incomplete immunity.
  - Recovery — Clinical improvement within 48 hours of rash; Occurrence of fever after D3 – 4 of rash suggests a measles complication.

1.2 COMPLICATIONS OF MEASLES

- Risk factors for complicated measles: Age < 5-years-old, pregnant, immunocompromised, vitamin A deficiency or malnourished

SYSTEM	COMPLICATIONS
Infection	Secondary infection & co-infection (bacterial and/or viral)
Gastrointestinal	Diarrhea (most common), gingivostomatitis, gastroenteritis, hepatitis
Pulmonary	Bronchopneumonia, laryngotracheobronchitis, bronchiolitis Co-infection: bacterial (strep pneumoniae, strep pyogenes, Hib, staph aureus), viral (parainfluenza, adenovirus, enterovirus, influenza, RSV)
CNS	Encephalitis within a few days of rash ~R5 [R1 – R14] Acute Disseminated Encephalomyelitis (ADEM) - During convalescent phase, within 2 weeks of rash - Clinical presentation: ◆ Fever, headache, meningism ◆ Altered metal state ◆ Multifocal neurological deficits (*Long tract signs, cerebellar ataxia, cranial neuropathies, bladder/bowel control, sensory deficits, myoclonus, choreoathetosis)*

1.3 MODIFIED MEASLES

- An attenuated infection that occurs in individuals with incomplete immunity. Examples:
  - Received post-exposure prophylaxis (MMR vaccine, Immunoglobulin)
  - Transplacental transfer of anti-measles antibodies
- Clinical manifestations are generally milder – with a longer incubation period (up to 28 days)

2. CLINICALLY IMPORTANT INFORMATION FOR EVALUATION

1. Date of birth / Age	
2. Past medical history	<u>Severely Immunocompromised</u>: 1. Severe primary immunodeficiency (PID) 2. Recipient of a bone marrow transplant (BMT) ▫ On immunosuppressive treatment, or within 12 months of completion of immunosuppressive treatment ▫ Complicated by graft-vs-host-disease (GVHD) 3. Patients on treatment for acute lymphoblastic leukemia (ALL) on immunosuppressive chemotherapy or within 6 months after completion of chemotherapy 4. Patients with diagnosis of AIDS or HIV-infected persons with: ▫ CD4 % < 15% (all ages), <u>or</u> ▫ CD4 count < 200 lymphocytes/mm^3 (age > 5 years), <u>or</u> ▫ Have not received MMR vaccine since receiving effective ART 5. Infants < 12 months whose mothers received Biologic Response Modifiers (BRMs)*; patients currently on, or received BRMs within the last 12 months. 6. Solid organ transplant recipients. 7. Patients who have received high dose corticosteroids (≥ 2 mg/kg/day or ≥ 20 mg/day of prednisolone or its equivalent for people who weigh ≥ 10kg) for ≥ 14 days within the past 4 weeks.
3. Immunization history	<u>Evidence of Immunity</u> 2 doses of MMR vaccination, both administered at age ≥ 12-months, separated by ≥ 28 days
4. Circumstances of exposure	▫ First and last day of exposure ▫ Was index case laboratory-confirmed? ▫ Presence of symptoms – URTI, rash ▫ Any direct contact? Any contact with contaminated surface?
5. Clinical signs & sx	<u>Clinical Description</u> 1. Generalized, maculopapular rash lasting ≥ 3 days; 2. Temperature ≥ 38.3 °C; 3. Cough, coryza & conjunctivitis

<u>Corticosteroids Equivalence Chart</u>

	Equivalent Dose (mg)
Prednisolone	5
Hydrocortisone	20
Methylprednisolone	4
Dexamethasone	0.75

**dependent on type of BRM mother received*

3. POST-EXPOSURE PROPHYLAXIS (PEP)

- Patients exposed to measles who cannot readily show that they have evidence of immunity should be offered PEP which can protect/prevent and modify the clinical course of measles.
- Options include MMR vaccination & Immune Globulin
- Refer to section 7 for further advice on subsequent implications on primary series of MMR vaccinations in infants.

	MMR VACCINE	IMMUNE GLOBULIN (IM/IV)
Window	Within 72 hours of measles exposure	Within 6 days of measles exposure
Indication	≥ 6-months-old Immunocompetent, **AND**	< 6-months-old regardless or maternal serostatus < 12-months-old & non-immune, **OR** Severely immunocompromised (any age)
Contra-indication	History of anaphylactic reactions to neomycin, **OR** History of severe allergic reaction to any component of the vaccine (Egg allergy is NOT a contraindication[1])	
Quarantine	21 days	28 days

3.1 POST-EXPOSURE QUARANTINE

- No PEP = 21 days
- MMR Vaccine PEP = 21 days
- IM/IVIG PEP = 28 days

4. CASE CLASSIFICATION

PROBABLE	CONFIRMED
- Meets clinical description, and - No epidemiological link to a laboratory confirmed measles case, and non-contributory or no measles laboratory testing	- Acute febrile illness with rash; and - Any of the following: o Isolation of measles virus from a clinical specimen; or o Detection of measles from a clinical specimen using PCR; or o IgG seroconversion or a significant rise in measles IgG antibody; or o A positive serologic test for measles IgM; or o Direct epidemiologic linkage to a case confirmed by one of the methods above

[1] MMR vaccine components are cultured in chick embryo fibroblast tissue culture.

5. MANAGEMENT OF SYMPTOMATIC PATIENTS

5.1 NOTIFY MOH

5.2 CONFIRMING MEASLES DIAGNOSIS

TEST	WINDOW	
Measles IgM	R3 to R30 (peak = R8 to R10)	*R0 = Day of onset of rash* *R3 = 3 days after onset of rash*
Measles IgG	R7 onward (peak = R14 to R16)	
Measles PCR	Highest yield = R0 to R3 but can be positive up to R10	

- Negative results from samples taken outside the window period may be false negative

5.3 DIFFERENTIAL DIAGNOSIS

- Viral illness with exanthems (e.g., adenovirus, enterovirus, rubella, EBV/CMV, influenza)
- Bacterial: Group A streptococcal infection (e.g., scarlet fever, toxic shock syndrome), Mycoplasma pneumoniae
- Kawasaki disease, Vasculitis (e.g., IgA vasculitis)

5.4 TREATMENT

- Patients should be placed in a single-patient airborne infection isolation room
- Airborne precautions should be taken by all HCW regardless of vaccination status

5.4.1 SUPPORTIVE THERAPY

- IV fluids, anti-pyretics
- Treat secondary bacterial infections (pneumonia, otitis media)

5.4.2 VITAMIN A SUPPLEMENTATION

- All children diagnosed with severe measles should receive Vitamin A supplementation (including patients who are not malnourished)
- Benefits:
  - Reduces measles deaths, severity & risk of complications
  - Prevents eye damage & blindness
- 2 doses of Vitamin A, given 24 hours apart
  - < 6 months 50 000 IU/day
  - 6 – 11 months 100 000 IU/day
  - ≥ 12 months 200 000 IU/day

6. DISPOSITION

6.1 ADMISSION

- Placed in a single airborne infection isolation room (NEP room)
- Indications
  - Clinically unwell & requires active supportive therapy (e.g., dehydration)
  - Presents with complicated measles (e.g., pneumonia, encephalitis)
  - High risk patients: immunocompromised, chronic disease with risk of decompensation

6.2 DISCHARGE

- Symptomatic medication
- Patients who received IM/IVIg should be given a memo with advice on quarantine duration & future MMR vaccination recommendations.
- Seek advice from Paeds ID regarding follow-up (to trace investigations and to assess patient for complications)

7. MEASLES VACCINATION FOR PREVENTION OF MEASLES

- Evidence of immunity is established = 2 doses of MMR, both administered at ≥ 12-months of age, separated by ≥ 28 days
- Patients who received PEP should still undergo measles vaccination (PEP received before 12-months are not counted)

	MMR VACCINE PEP	IMIG/IVIG PEP
MMR immunization	Patients aged ≥ 6 months to < 12 months require 2 doses of MMR, both after age ≥ 12 months - Dose #1 @ ≥ 12-months-old (at least ≥ 28 days after PEP MMR vaccine) - Dose #2 @ 28 days after Dose #1 Patients aged ≥ 12 months require 1 dose of MMR @ ≥ 28 days after PEP MMR vaccine	Patients who received IGIM/IV measles PEP will need 2 doses to complete their measles primary vaccination series. Their 1st dose should be at least: - 6 months (IM) later - 8 months (IV) later

8. SUMMARY DIAGRAM FOR PATIENTS ELIGIBLE FOR PEP & SUBSEQUENT FOLLOW UP